

Supplementary File for Learning Dynamic Coupling of Neural Oscillations for Motor Imagery Classification

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This supplementary file consists of the following parts:

- 1) **Notations and Acronyms**
- 2) **Optimization of CFC-based Tensor Decomposition**
- 3) **Cross-frequency Coupling Operations**
- 4) **r^2 -value Operations**

I. NOTATIONS AND ACRONYMS

TABLE I
SUMMARY OF MATHEMATICAL SYMBOLS USED IN THE PAPER.

Symbol	Meaning	First Appearance
$\mathcal{X}, \mathcal{Y}, \mathcal{Z}$	Tensors (order- D arrays)	Section II.A
$\mathbf{X}, \mathbf{Y}, \mathbf{Z}$	Matrices	Section II.A
$\mathbf{x}, \mathbf{y}, \mathbf{z}$	Vectors	Section II.A
x, y, z	Scalars	Section II.A
$\mathbb{R}$	Real number set	Section II.A
I_d	Size of tensor mode- d	Section II.A
$\times_d$	Mode- d product	Eq. (1)
$ \cdot $	Absolute value / Frobenius norm	Eq. (4)
$\arg$	Argument of complex number	Eq. (4)
$\log$	Natural logarithm	Eq. (10)
$H(\cdot)$	Entropy	Eq. (12)
$I(\cdot; \cdot)$	Mutual information	Eq. (11)
$\psi(t, f)$	Complex Morlet wavelet	Eq. (2)
f_b	Wavelet bandwidth parameter	Eq. (2)
K	Number of EEG channels	Section II.B
N	Number of frequency bins	Section II.B
L	Number of time samples	Section II.B
M	Total number of spatial-spectral factors	Section II.C
M_s	Number of selected subject-specific factors	Section II.D
r^2	Discrimination metric (r -squared)	Section V.E
p	Statistical p -value	Section V.E
κ	Cohen's kappa coefficient	Section V.C

Table I presents a nomenclature of all mathematical symbols used in this paper, together with their meanings and specifying the equations or sections where they first appear. Table II lists all abbreviations (e.g., BCI, MI, EEG, CFC, CTD, AAC, PPC, PAC, CWT, FBCSP, CNN, and LRP) alongside their corresponding full forms.

TABLE II
SUMMARY OF ACRONYMS USED IN THE PAPER.

Acronym	Full Form
AAC	Amplitude-Amplitude Coupling
ADFCNN	Attention-based Dual-scale Fusion CNN
BCI	Brain-Computer Interface
CFC	Cross-Frequency Coupling
CFCNet	Cross-Frequency Coupling Network
CNN	Convolutional Neural Network
CSP	Common Spatial Patterns
CTD	CFC-based Tensor Decomposition
CTNet	Convolutional Transformer Network
CWT	Continuous Wavelet Transform
DL	Deep Learning
ERD	Event-Related Desynchronization
ERS	Event-Related Synchronization
FBCSP	Filter Bank Common Spatial Patterns
IFNet	Interactive Frequency Convolutional Neural Network
LRP	Layer-wise Relevance Propagation
MI	Motor Imagery
PAC	Phase-Amplitude Coupling
PCC	Pearson Correlation Coefficient
PPC	Phase-Phase Coupling
SMR	Sensorimotor Rhythm

II. OPTIMIZATION OF CFC-BASED TENSOR DECOMPOSITION (CTD)

This section describes the optimized detail for the proposed CFC-based Tensor Decomposition (CTD) used in CFCNet. To ensure that the extractive features are discriminative, we employ the CSP technique to find the parameters of $\mathbf{s}$ and $\mathbf{B}$. To be more specific, the optimization objective of CTD is to maximize the following power ratios for class 1 and class 2:

$$\max_{\mathbf{s}, \mathbf{B}} \frac{E_{\mathcal{X} \in C_1}[\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})]}{E_{\mathcal{X} \in C_2}[\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})] + E_{\mathcal{X} \in C_1}[\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})]} \quad (1)$$

Here, C_y is the set in training data, which contains the test belonging to class $y \in \{1, 2\}$; $\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B}) = \mathbf{z}^T \mathbf{z}$ denotes the power calculated from the projection signal $\mathbf{z}$, i.e., $\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B}) = \mathbf{z}^T \mathbf{z}$; and $E_{\mathcal{X} \in C_y}[\cdot]$ denotes the expected covariance matrix over C_y . Since it is difficult to determine $\mathbf{s}$ and $\mathbf{B}$ simultaneously, a standard alternating optimization strategy is adopted in the optimization process.

While $\mathbf{B}$ is fixed, $\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})$ can be denoted by $\mathbf{s}^T \mathbf{G} \mathbf{G}^T \mathbf{s}$, where $\mathbf{G} = \mathcal{X} \times_2 [\mathbf{B}^T] \in \mathbb{R}^{K \times L}$. Let $\text{cov}(\mathbf{A}, y)$ be the function that computes the average covariance matrix for class y . Then it follows that $E_{\mathcal{X} \in C_1}[\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})] = \mathbf{s}^T \text{cov}(\mathbf{G}, 1) \mathbf{s}$ and $E_{\mathcal{X} \in C_2}[\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B})] = \mathbf{s}^T \text{cov}(\mathbf{G}, 2) \mathbf{s}$. Thus, (1) can be re-expressed as a Rayleigh quotient maximization problem.

$$\max_{\mathbf{s}_m} J(\mathbf{s}_m | \mathbf{B}_m) = \frac{\mathbf{s}_m^T \text{cov}(\mathbf{G}_m, 1) \mathbf{s}_m}{\mathbf{s}_m^T [\text{cov}(\mathbf{G}_m, 1) + \text{cov}(\mathbf{G}_m, 2)] \mathbf{s}_m} \quad (2)$$

Problem (2) can be solved by generalized eigenvalue decomposition:

$$\text{cov}(\mathbf{G}_m, 1) \mathbf{s}_m = \lambda [\text{cov}(\mathbf{G}_m, 1) + \text{cov}(\mathbf{G}_m, 2)] \mathbf{s}_m \quad (3)$$

The spatial factor $\mathbf{s}$ is given by the eigenvector corresponding to the eigenvalue λ . The largest or smallest eigenvalues mean discriminability of the eigenvectors. We set M_s as an even number $2r$. And then half of the factors, $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_r\}$, are chosen as the eigenvectors corresponding to the largest r eigenvalues for class 1, and the other factors, $\{\mathbf{s}_{r+1}, \mathbf{s}_{r+2}, \dots, \mathbf{s}_{2r}\}$ are the eigenvectors corresponding to the smallest r eigenvalues for class 2.

Next, we find $\{\mathbf{B}_m, m = 1, 2, \dots, M\}$ assuming that $\{\mathbf{s}_m, m = 1, 2, \dots, M\}$ is given. Let $\beta_m = \text{vec}(\mathbf{B}_m) \in \mathbb{R}^{KN}$, where $\text{vec}(\mathbf{B}_m)$ denotes the vectorization operation that sequentially concatenates the column vectors of $\mathbf{B}_m$ to form a new vector.

Algorithm 1 Optimization details for CTD

Input: The training set $\{\mathbf{X}_{i\text{-th}} \in \mathbb{R}^{K \times L}, i = 1, 2, \dots, I\}$ with their labels $\mathbf{y} \in \mathbb{R}^I$, the maximum number of iterations T and a small value ε for testing convergence, and the number of spatial-spectral factors $2r$.

Output: $\{(\mathbf{s}_m, \mathbf{B}_m), m = 1, 2, \dots, M\}$ (note that $M = 2r$).

calculate $\mathcal{X}_i \in \mathbb{R}^{K \times N \times L}$ for $1 \leq i \leq I$;

calculate average covariance: $\text{cov}(\mathbf{R}, 1)$ and $\text{cov}(\mathbf{R}, 2)$;

calculate $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{2r}\}$ according to (7);

initialize $\{\mathbf{B}_m, m = 1, 2, \dots, 2r\}$ according to (8);

for $1 \leq t \leq T$ **do**

 calculate $\text{cov}(\mathbf{G}_m, 1)$ and $\text{cov}(\mathbf{G}_m, 2)$;

 obtain $\{\mathbf{s}_m^{(t)}, m = 1, 2, \dots, 2r\}$ by solving (2);

 calculate $\text{cov}(\mathbf{Q}_m, 1)$ and $\text{cov}(\mathbf{Q}_m, 2)$;

 obtain $\{\mathbf{b}_{m,k}^{(t)}, k = 1, 2, \dots, K, m = 1, 2, \dots, 2r\}$ by solving (4) and (5);

 calculate $\{\mathbf{B}_m^{(t)}, m = 1, 2, \dots, 2r\}$ according to (8);

 if $t = T$ or $J(\mathbf{s}_m^{(t)} | \mathbf{B}_m^{(t-1)}) - J(\mathbf{B}_m^{(t)} | \mathbf{s}_m^{(t)}) \leq \varepsilon$ for all m

 break;

end for

return $\{(\mathbf{s}_m^{(t)}, \mathbf{B}_m^{(t)}), m = 1, 2, \dots, M\}$

Then it follows that $\alpha(\mathcal{X}; \mathbf{s}_m, \mathbf{B}_m) = \beta_m^T \mathbf{Q}_m \mathbf{Q}_m^T \beta_m$, where the matrix $\mathbf{Q}_m = [s_{m,1} \mathbf{X}_1^T \ s_{m,2} \mathbf{X}_2^T \ \dots \ s_{m,K} \mathbf{X}_K^T]^T \in \mathbb{R}^{(KN) \times L}$, with $s_{m,k}$ being the k -th element of $\mathbf{s}_m$. Similarly, the optimization problem can be written as

$$\max_{\mathbf{B}_m} J(\mathbf{B}_m | \mathbf{s}_m) = \frac{\beta_m^T \text{cov}(\mathbf{Q}_m, 1) \beta_m}{\beta_m^T [\text{cov}(\mathbf{Q}_m, 1) + \text{cov}(\mathbf{Q}_m, 2)] \beta_m} \quad (4)$$

In addition, problem (4) is solved by generalized eigenvalue decomposition. β_m is chosen as the eigenvector corresponding to the largest eigenvalue for m within the range of 1 to r , and the eigenvector corresponding to the smallest eigenvalue for m within the range of $r + 1$ to $2r$. The k -th column of $\mathbf{B}_m$, $\mathbf{b}_{m,k}$, can be restored from β_m by

$$\mathbf{b}_{m,k} = [\beta_{m,p} \ \beta_{m,p+1} \ \dots \ \beta_{m,p+N-1}]^T, \quad k = 1, 2, \dots, K, \ m = 1, 2, \dots, 2r \quad (5)$$

where $p = (k - 1)N + 1$, and $\beta_{m,p}$ is the p -th element of β_m .

To facilitate fast convergence, we initialize $\{\mathbf{B}_m, m = 1, 2, \dots, M\}$ with effective initial values. Let us define a coupling factor $\mathbf{w} \in \mathbb{R}^{KN}$ as

$$\mathbf{w} = \mathbf{s} \odot \text{vec}(\mathbf{B}) = [s_1 \mathbf{b}_1^T \ s_2 \mathbf{b}_2^T \ \dots \ s_K \mathbf{b}_K^T]^T \quad (6)$$

where $\odot$ denotes the KhatriRao product of two vectors. It is easy to find $\alpha(\mathcal{X}; \mathbf{s}, \mathbf{B}) = \mathbf{w}^T \mathbf{R} \mathbf{R}^T \mathbf{w}$, where $\mathbf{R} = [\mathbf{X}_1; \mathbf{X}_2; \dots; \mathbf{X}_K] \in \mathbb{R}^{(KN) \times L}$. Thus, (1) can be re-expressed as

$$\max_{\mathbf{w}} J(\mathbf{w}) = \frac{\mathbf{w}^T \text{cov}(\mathbf{R}, 1) \mathbf{w}}{\mathbf{w}^T [\text{cov}(\mathbf{R}, 1) + \text{cov}(\mathbf{R}, 2)] \mathbf{w}} \quad (7)$$

Problem (7) also can be solved by generalized eigenvalue decomposition: $\gamma[\text{cov}(\mathbf{R}, 1) + \text{cov}(\mathbf{R}, 2)] \mathbf{w}$. Half of the factors, $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r\}$, are chosen as eigenvectors corresponding to the largest r eigenvalues for class 1, and the others are eigenvectors corresponding to the smallest r eigenvalues for class 2. The initial values of $\mathbf{B}_m$ can be estimated from $\mathbf{w}_m$, as follows:

$$\mathbf{b}_{m,k} = [w_{m,p} \ w_{m,p+1} \ \dots \ w_{m,p+N-1}]^T / h_p, \quad k = 1, 2, \dots, K, \ m = 1, 2, \dots, 2r \quad (8)$$

where $p = (k - 1)N + 1$; $w_{m,p}$ is the p -th element of $\mathbf{w}_m$; and $h_p = \sqrt{(w_{m,p})^2 + (w_{m,p+1})^2 + \dots + (w_{m,p+N-1})^2}$. In this way, $\mathbf{b}_{m,k}$ is assumed to be unit l_2 -norm. The pseudo-code of the CTD finding $\{(\mathbf{s}_m, \mathbf{B}_m), m = 1, 2, \dots, M\}$ is given in **Algorithm 1**. Similarly to CSP, CTD can easily be extended to multi-class applications, by adopting strategies, such as Pair-Wise and One-Versus-Rest.

The proposed CTD method finds spatial-spectral factors alternatively by solving a single Rayleigh quotient maximization objective function, where we never decrease the value of cost function between two successive iterations. The convergence of the alternating optimization process is proved in *Theorem 1*.

Theorem 1: The alternating optimization process for CTD guarantees monotonic convergence:

$$\begin{aligned} \sigma &\leq J(\mathbf{s}_m^{(1)}|\mathbf{B}_m^{(0)}) \leq J(\mathbf{B}_m^{(1)}|\mathbf{s}_m^{(1)}) \leq J(\mathbf{s}_m^{(2)}|\mathbf{B}_m^{(1)}) \leq \dots \\ &\leq J(\mathbf{s}_m^{(t)}|\mathbf{B}_m^{(t-1)}) \leq J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)}) < 1 \end{aligned} \quad (9)$$

where σ is a limiting value in the $\mathbb{R}^+$ space ($0 < \sigma < 1$), and $J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)})$ is calculated with the given $\mathbf{s}_m^{(t)}$ in the t -th ($t \geq 1$) iteration. $\mathbf{B}_m^{(0)}$ is initialized according to (8).

Proof: Let $\sigma_m = J(\mathbf{w}_m)$ for $m = 1, 2, \dots, 2r$, and it is easy to know that $0 < \sigma_m < 1$. Since $\mathbf{s}_m^{(1)}$ is calculated by arg maximizing $J(\mathbf{s}_m|\mathbf{B}_m)$ with the given $\mathbf{B}_m^{(0)}$, we have the following inequality: $\sigma \leq J(\mathbf{s}_m^{(1)}|\mathbf{B}_m^{(0)})$, where $\sigma = \min(\sigma_1, \sigma_2, \dots, \sigma_{2r})$. According to (4), we have $J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)}) < 1$. In addition, because $\mathbf{s}_m^{(t)}$ maximizes $J(\mathbf{s}_m^{(t)}|\mathbf{B}_m^{(t-1)})$ with fixed $\mathbf{B}_m^{(t-1)}$ and then $\mathbf{B}_m^{(t)}$ maximizes $J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)})$ with calculated $\mathbf{s}_m^{(t)}$, the relationship $J(\mathbf{B}_m^{(t-1)}|\mathbf{s}_m^{(t-1)}) \leq J(\mathbf{s}_m^{(t)}|\mathbf{B}_m^{(t-1)}) \leq J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)})$ is given. Hence, we finally obtain the following relationship: $\sigma \leq J(\mathbf{s}_m^{(1)}|\mathbf{B}_m^{(0)}) \leq J(\mathbf{B}_m^{(1)}|\mathbf{s}_m^{(1)}) \leq \dots \leq J(\mathbf{s}_m^{(t)}|\mathbf{B}_m^{(t-1)}) \leq J(\mathbf{B}_m^{(t)}|\mathbf{s}_m^{(t)}) < 1$.

III. CROSS-FREQUENCY COUPLING OPERATIONS

PHASE-AMPLITUDE COUPLING VIA KULLBACK-LEIBLER MODULATION INDEX (KL-MI)

The Kullback-Leibler Modulation Index (KL-MI) was employed to quantify the Phase-Amplitude Coupling (PAC) between low-frequency phase and high-frequency amplitude across C EEG channels. Let $x_c(t)$ denote the univariate EEG time series for channel $c \in \{1, \dots, C\}$ at time t . The computational pipeline is defined as follows:

1. Signal Filtering and Feature Extraction

The raw signal $x_c(t)$ was band-pass filtered into a low-frequency component (phase-giving, f_{low}) and a high-frequency component (amplitude-giving, f_{high}):

$$x_{\text{low}}(t) = \text{BandPass}(x_c(t); f_{\text{low}}), \quad x_{\text{high}}(t) = \text{BandPass}(x_c(t); f_{\text{high}}). \quad (10)$$

The analytic signals were obtained via the Hilbert transform $\mathcal{H}\{\cdot\}$:

$$z_{\text{low}}(t) = x_{\text{low}}(t) + i\mathcal{H}\{x_{\text{low}}(t)\}, \quad z_{\text{high}}(t) = x_{\text{high}}(t) + i\mathcal{H}\{x_{\text{high}}(t)\}. \quad (11)$$

The instantaneous phase $\phi(t)$ and amplitude envelope $a(t)$ were extracted as:

$$\phi(t) = \arg(z_{\text{low}}(t)), \quad a(t) = |z_{\text{high}}(t)|. \quad (12)$$

2. Probability Distribution Estimation

The phase range $[0, 2\pi)$ was partitioned into K discrete bins B_k :

$$B_k = \left[\frac{2\pi(k-1)}{K}, \frac{2\pi k}{K} \right), \quad k = 1, \dots, K. \quad (13)$$

Simultaneously, the amplitude range $[0, A_{\text{max}}]$ was partitioned into L bins. The conditional probability $f_{k,l}$ of observing amplitude in bin l given the phase in bin k , and the marginal probability g_l of amplitude across all time points, were estimated via histogram normalization:

$$f_{k,l} = \frac{\#\{t : \phi(t) \in B_k \text{ and } a(t) \in \text{bin } l\}}{\#\{t : \phi(t) \in B_k\}}, \quad (14)$$

$$g_l = \frac{\#\{t : a(t) \in \text{bin } l\}}{T}. \quad (15)$$

3. KL Divergence and Modulation Index

The Kullback-Leibler (KL) divergence $D_{\text{KL}}(k)$ between the conditional distribution f_k and the marginal distribution g was computed for each phase bin k :

$$D_{\text{KL}}(k) = \sum_{l=1}^L f_{k,l} \log_2 \frac{f_{k,l}}{g_l}. \quad (16)$$

The mean KL divergence across all phase bins is:

$$\bar{D}_{\text{KL}} = \frac{1}{K} \sum_{k=1}^K D_{\text{KL}}(k). \quad (17)$$

Finally, the KL-MI was normalized by $\log_2 K$ to yield a value within $[0, 1]$:

$$\text{KL-MI} = \frac{\overline{D}_{\text{KL}}}{\log_2 K}. \quad (18)$$

A value of 0 indicates no coupling, whereas 1 indicates maximal phase-amplitude coupling.

4. Statistical Significance

To assess statistical significance, a permutation test was performed by randomly shuffling the phase time series relative to the amplitude series (1000 permutations) to generate a null distribution of KL-MI values. The p -value was calculated as the proportion of permuted values exceeding the observed KL-MI.

A. Phase-Amplitude Coupling

PAC [1] describes the modulation relationship between the phase of low-frequency oscillations and the amplitude of high-frequency oscillations. We employed the modulation index based on Kullback-Leibler (KL-MI) [2] to measure the PAC. The KL-MI is calculated as follows:

1) For CTD, let $\mathbf{B}_p$ and $\mathbf{B}_A$ be the spectral factors related to phase and amplitude, respectively. Both of them correspond to the highest M_s mutual information. The phase and amplitude representations, denoted as $\mathbf{G}_p \in \mathbb{R}^{K \times L}$ and $\mathbf{G}_A \in \mathbb{R}^{K \times L}$, are extracted from two trials of different MI tasks in the training set. These representations are obtained by filtering the trial signals using (6) in the paper with $\mathbf{B}_p$ and $\mathbf{B}_A$, respectively. Please refer to Section II.C of the paper for the operational specifics.

2) Let $\mathbf{g}_p \in \mathbb{R}^L$ and $\mathbf{g}_a \in \mathbb{R}^L$ be a channel signal extracted from $\mathbf{G}_p$ and $\mathbf{G}_A$, respectively. The phase range of $\mathbf{g}_p$ is partitioned into H (typically $H = 18$) discrete bins B_h :

$$B_h = \left[\frac{2\pi(h-1)}{H}, \frac{2\pi h}{H} \right], \quad h = 1, \dots, H \quad (19)$$

Simultaneously, the amplitude range of $\mathbf{g}_a$ is partitioned into S bins, i.e., A_s , $s = 1, \dots, S$.

3) The conditional probability $p_{h,s}$ of observing amplitude in bin s given the phase in bin h , and the marginal probability p_s of amplitude across all time points, are estimated as follows:

$$p_{h,s} = \frac{\sum_{l=1}^L \mathbf{1}(\mathbf{g}_p \in B_h) \cdot \mathbf{1}(\mathbf{g}_a \in A_s)}{\sum_{l=1}^L \mathbf{1}(\phi(t) \in B_h)} \quad (20)$$

$$p_s = \frac{1}{L} \sum_{l=1}^L \mathbf{1}(\mathbf{g}_a \in A_s) \quad (21)$$

Here, $\mathbf{1}(P)$ equals 1 if proposition P is true, and 0 otherwise.

4) The Kullback-Leibler (KL) divergence $D_{\text{KL}}(h)$ between the conditional distribution p_h and the marginal distribution p_l is computed for each phase bin h :

$$D_{\text{KL}}(h) = \sum_{s=1}^S p_{h,s} \log_2 \frac{p_{h,s}}{p_s} \quad (22)$$

The mean KL divergence across all phase bins is:

$$\overline{D}_{\text{KL}} = \frac{1}{H} \sum_{h=1}^H D_{\text{KL}}(h) \quad (23)$$

5) Finally, the KL-MI is normalized by $\log_2 H$ to yield a PAC value within $[0, 1]$:

$$\text{KL-MI} = \frac{\overline{D}_{\text{KL}}}{\log_2 H} \quad (24)$$

A value of 0 indicates no coupling, whereas 1 indicates maximal phase-amplitude coupling.

B. Phase-Phase Coupling

We employ phase-locking value (PLV) [4] to calculate the PPC coefficient. Let $\mathbf{g}_{f_1} \in \mathbb{R}^L$ and $\mathbf{g}_{f_2} \in \mathbb{R}^L$ denote two channel signals that located in centre frequencies f_1 and f_2 , respectively, extracted from the phase representations of two trials. The PLV value to quantify f_1 - f_2 phase-phase coupling between $\mathbf{g}_{f_1}$ and $\mathbf{g}_{f_2}$ is defined as follows

$$\text{PLV} = \frac{1}{L} \left| \sum_{t=1}^L e^{j(\Delta\phi(f_1, f_2, t))} \right| \quad (25)$$

where $\Delta\phi(f_1, f_2, t)$ denotes the t -point phase difference between $\mathbf{g}_{f_1}$ and $\mathbf{g}_{f_2}$. It is calculated as follows:

$$\Delta\phi(f_1, f_2, t) = g_{f_1, t} - g_{f_2, t} \quad (26)$$

In (26), $g_{f_1, t}$ and $g_{f_2, t}$ are the signals $\mathbf{g}_{f_1}$ and $\mathbf{g}_{f_2}$ at time t , respectively.

In the paper, we extract five rhythms (frequency bands) for the PPC analysis, i.e., θ (4–8 Hz), α (8–12 Hz), β_1 (12–20 Hz), β_2 (20–30 Hz), and γ (30–40 Hz). Thus the centre frequencies a and b can be decided according to the extracted rhythms.

C. Amplitude-Amplitude Coupling

We employ the Pearson correlation coefficient (PCC) [4] to explore the AAC relationship of signals. Let $\mathbf{g}_{k1} \in \mathbb{R}^L$ and $\mathbf{g}_{k2} \in \mathbb{R}^L$ denote the vectorized signals from channels $k1$ and $k2$, respectively, extracted from the amplitude representations of two trials. The PCC value to quantify AAC between $\mathbf{g}_{k1}$ and $\mathbf{g}_{k2}$ is defined as follows

$$\text{PCC} = \frac{\sum_{t=1}^{I_2 L} (g_{k1, t} - \bar{\mathbf{g}}_{k1})(g_{k2, t} - \bar{\mathbf{g}}_{k2})}{\sqrt{\sum_{t=1}^{I_1 L} (g_{k1, t} - \bar{\mathbf{g}}_{k1})^2} \sqrt{\sum_{t=1}^{I_2 L} (g_{k2, t} - \bar{\mathbf{g}}_{k2})^2}} \quad (27)$$

In (27), $g_{k1, t}$ and $g_{k2, t}$ are the signals $\mathbf{g}_{k1}$ and $\mathbf{g}_{k2}$ at time t , respectively.

IV. r^2 -VALUE OPERATIONS

The r^2 -value [5] is used to measure the separability of the features obtained from EEG signals, which can be calculate as follows:

$$r^2 = \left(\frac{\sqrt{I_1 * I_2} \text{mean}\{F_i | e_i = 1\} - \text{mean}\{F_i | e_i = 2\}}{I_1 + I_2 \text{std}\{F_i | c_i = 1, 2\}} \right) \quad (28)$$

where I_1 and I_2 denote the number of trials with two categories, respectively. e_i is the category of i -th trial. F_i is the log-power feature of the i -th trial. A higher r^2 -value indicates a higher separability of the EEG signal features in terms of their categories.

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